Infinite-Horizon MDPs

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$$\min_{\pi} E\left(\sum_{k=0}^{\infty} \alpha^k r(s_k, a_k)\right) \text{ with } s_0 \sim \rho$$

where $\alpha \in [0,1)$ is the discount factor,

$\pi = (\pi_0, \pi_1, \pi_2, \ldots)$, where

$a_k \sim \pi_k(\cdot \mid H_k)$ (π_k is a distribution over actions which could depend on the history $H_k = \{s_0, a_0, s_1, a_1, \ldots, s_{k-1}, a_{k-1}, s_k)$

ρ : initial state distribution

Note: Here, we assume $r(s,a)$ is deterministic given (s,a). We can also allow $r(s,a)$ to be a random variable, in which case we have to replace $r(s,a)$ by $\bar{r}(s,a) = E(r(s,a) \mid s,a)$.

- We first consider stationary, Markov policies and later we will show that this class of policies are sufficient to consider, i.e.,

$$a_k \sim \pi(\cdot \mid s_k)$$

no explicit dependence on k

depends only on s_k and not the entire history H_k.

Fixed stationary policy π (possibly randomized):

Define the value function V as

$$V_\pi(s) = E\left(\sum_{k=0}^{\infty} \alpha^k r(s_k, a_k) \,\middle|\, s_0 = s\right),$$

with $a_k \sim \pi(\cdot \mid s_k)$

This sum exists since $\{s_k, a_k\} \in$ finite sets and hence $\max_{s,a} |r(s,a)| = r_{max}$ exists and therefore,

$$\left|\sum_{k=0}^{\infty} \alpha^k r(s_k, a_k)\right| \leq \frac{r_{max}}{1-\alpha} \quad a.s.$$

Note that

$$V_\pi(s) = E\left(r(s_0, a_0) + \sum_{k=1}^{\infty} \alpha^k r(s_k, a_k) \,\middle|\, s_0 = s\right)$$

$$= E(r(s_0, a_0) \mid s_0 = s) + E\left(\sum_{k=1}^{\infty} \alpha^k r(s_k, a_k) \,\middle|\, s_0 = s\right)$$

$$= E(r(s_0, a_0) \mid s_0 = s) + \alpha \underbrace{E\left(\sum_{k=1}^{\infty} \alpha^{k-1} r(s_k, a_k) \,\middle|\, s_0 = s\right)}$$

$$= E\left(E\left(\sum_{k=1}^{\infty} \alpha^{k-1} r(s_k, a_k) \,\middle|\, s_1, s_0 = s\right) \,\middle|\, s_0 = s\right)$$

$$\overset{\text{Markov}}{=} E\left(E\left(\sum_{k=1}^{\infty} \alpha^{k-1} r(s_k, a_k) \,\middle|\, s_1\right) \,\middle|\, s_0 = s\right)$$

$$= E(V_\pi(s_1) \mid s_0 = s)$$

$$\Rightarrow \boxed{V_\pi(s) = r(s, \pi) + \alpha E(V_\pi(s_1) \mid s_0 = s)} \quad — ①$$

Here, $r(s, \pi)$ is short-hand notation for

$$\sum_a r(s,a) \pi(a \mid s)$$

and $E(V_\pi(s_1) \mid s_0 = s) = \sum_{s_1} P(s_1 \mid s_0 = s, a) \pi(a \mid s) V_\pi(s_1)$

Define the vector $V_\pi = \begin{pmatrix} V_\pi(1) \\ \vdots \\ V_\pi(|S|) \end{pmatrix}$,

where $|S|$ is the cardinality of the state space S.
Then, ① can be written as

$$\boxed{V_\pi = T_\pi(V_\pi)}, \quad \text{where } T_\pi : \mathbb{R}^{|S|} \to \mathbb{R}^{|S|}$$

with

$$T_\pi(V_\pi)(s) = r(s, \pi) + \alpha E(V_\pi(s_1) \mid s_0 = s)$$

Properties of T_π:

(i) T_π is an α-contraction in $\|\cdot\|_\infty$

(ii) If $V \leq \tilde{V}$, then $T_\pi V \leq T_\pi \tilde{V}$, where the inequalities should be interpreted as element by element. (Monotonicity)

Proof: (a) $T_\pi(V)(s) - T_\pi(\tilde{V})(s)$

$$= \cancel{r(s,\pi)} + \alpha E(V(s_1) \mid s_0 = s) - \cancel{r(s,\pi)} - \alpha E(\tilde{V}(s_1) \mid s_0 = s)$$

$$= \alpha \sum_{s_1} \sum_a \pi(a \mid s) P(s_1 \mid s, a) (V(s_1) - \tilde{V}(s_1))$$

$$\leq \alpha \|V - \tilde{V}\|_\infty \underbrace{\sum_a \pi(a \mid s) \underbrace{\sum_{s_1} P(s_1 \mid s, a)}_{=1}}_{=1}$$

$$\Rightarrow |T_\pi(V)(s) - T_\pi(\tilde{V})(s)| \leq \alpha \|V - \tilde{V}\|_\infty$$

$$\Rightarrow \boxed{\|T_\pi(V) - T_\pi(\tilde{V})\|_\infty \leq \alpha \|V - \tilde{V}\|_\infty}$$

(b) $T_\pi(V)(s) = r(s,\pi) + \alpha \sum_{s_1, a} P(s_1 \mid s, a) \pi(a \mid s) V(s_1)$

$$\leq r(s,\pi) + \alpha \sum_{s_1, a} P(s_1 \mid s, a)) \pi(a \mid s) \tilde{V}(s_1)$$

$$= T_\pi(\tilde{V})(s) \quad \forall s$$

$$\Rightarrow \boxed{T_\pi(V) \leq T_\pi(\tilde{V})}$$

□

- T_π is called the Bellman operator for a fixed policy π or the Bellman consistency operator.

- From the contraction property of the Bellman operator, it is clear that V_π can be obtained as the limit of the fixed-point iteration:

$$\boxed{V_{k+1} = T_\pi(V_k) \quad \text{for any } V_0.}$$

- Note that the Bellman operator can also be written as

$$T_\pi(V) = r_\pi + \alpha P_\pi V, \quad \text{where}$$

r_π is a vector whose i^{th} element is

$$r_\pi(i) = \sum_a r(i,a) \pi(a \mid i)$$

and P_π is a matrix whose $(i,j)^{th}$ element is

$$(P_\pi)_{ij} = \sum_a \pi(a \mid i) P(j \mid i, a). \text{ Thus,}$$

$V_\pi = T_\pi(P_\pi)$ can be written as

$$V_\pi = r_\pi + \alpha P_\pi V_\pi$$

and we can solve for V_π as

$$\boxed{V_\pi = (I - \alpha P_\pi)^{-1} r_\pi}$$

$(I - \alpha P_\pi)$ is invertible because all eigenvalues of αP_π have absolute value < 1 since $\alpha < 1$ by the Perron-Frobenius theorem for stochastic matrices, which we will accept without a proof here.

Bellman optimality equation: Suppose we know that a deterministic, stationary policy is optimal. Then, as in the case of finite-horizon problems, a reasonable guess for the Bellman equation is

$$\boxed{V(s) = \max_a r(s,a) + \alpha E(V(s_1) \mid s_0 = s, a_0 = a)}$$

where V is the optimal value function. If V is known, notice that the RHS returns a deterministic policy. We will show that this is indeed the optimal policy.

Before we do that, we write the Bellman equation as

$$V = T(V), \text{ where}$$

$$T(V)(i) = \max_a r(i,a) + \alpha E(V(s_1) \mid s_0 = i, a_0 = a)$$

The operator T is called the Bellman operator.

Properties of T:

(i) T is an α-contraction in $\|\cdot\|_\infty$

(ii) T is monotone: $TV \leq T\tilde{V}$ if $V \leq \tilde{V}$

Proof:

(i) $T(V)(s) - T(\tilde{V})(s)$

$$= \max_a r(s,a) + \alpha E(V(s_1) \mid s_0 = s) \leftarrow \text{let } a^* \text{ be the max here}$$

$$- \max_{a'} r(s,a') + \alpha E(\tilde{V}(s_1) \mid s_0 = s)$$

$$\leq \cancel{r(s,a^*)} + \alpha E(V(s_1) \mid s_0 = s) - \cancel{r(s,a^*)} + \alpha E(\tilde{V}(s_1) \mid s_0 = s)$$

$$= \alpha E(V(s_1) - \tilde{V}(s_1) \mid s_0 = s) \leq \alpha \|V - \tilde{V}\|_\infty$$

$$\Rightarrow \boxed{\|T(V) - T(\tilde{V})\|_\infty \leq \alpha \|V - \tilde{V}\|_\infty}$$

(b) Easy.

□

(a) $\Rightarrow$ There exists a unique V s.t. $V = T(V)$ and can be obtained as the limit of a fixed-point iteration.

Let π^* be the policy obtained by solving

$$\max_a r(s,a) + \alpha E(V(s_1) \mid s_0 = s, a_0 = a)$$

→ this gives us $\pi^*(s)$.

Note that π^* satisfies

$$V(s) = \underbrace{r(s, \pi^*(s)) + \alpha E(V(s) \mid s, \pi^*(s))}_{T_{\pi^*}(V)(s)}$$

$$\Leftrightarrow V = T_{\pi^*}(V)$$

Thus, V is also the value function of π^*

□

Theorem: π^* is optimal in the class of all policies.

Proof:

Step 1: Consider the problem of maximizing

$$E\left(\sum_{k=0}^{N-1} \alpha^k r(s_k, a_k)\right)$$

The value function satisfies

$$W_N(s_N) = 0$$

$$W_{N-1}(s) = \max_a r(s,a) + \alpha E(W_N(s_N) \mid s_{N-1} = s, a_{N-1} = a)$$

$$= T(0)(s)$$

$$\Leftrightarrow W_{N-1} = T(0)$$

$$W_{N-2}(s) = \max_a r(s,a) + \alpha E(W_{N-1}(s_{N-1}) \mid s_{N-2} = s, a_{N-2} = a)$$

$$= T^2(0)$$

$$\Leftrightarrow W_{N-2} = T^2(0)$$

Continuing, $W_0 = T^N(0)$

If π is an optimal policy (not necessarily Markovian deterministic), then

$$V_\pi \geq W_0 = T^N(0)$$

Taking the limit as $N \to \infty$,

$V_\pi \geq V$ (RHS is by the contraction mapping theorem)

Next,

$$V_\pi(s) = \max_\pi E\left(\sum_{k=0}^{\infty} \alpha^k r(s_k, a_k) \,\middle|\, s_0 = s\right)$$

$$\leq \max_\pi E\left(\sum_{k=0}^{N-1} \alpha^k r(s_k, a_k) \,\middle|\, s_0 = s\right) + \sum_{k=N}^{\infty} \alpha^k r_{max}$$

$$\leq \max_\pi E\left(\sum_{k=0}^{N-1} \alpha^k r(s_k, a_k) \,\middle|\, s_0 = s\right) + \frac{\alpha^N r_{max}}{1-\alpha}$$

$$= T^N(0) + \frac{\alpha^N r_{max}}{1-\alpha}$$

$$\to V(s) \text{ as } N \to \infty$$

$$\Rightarrow V_\pi = V \quad \Rightarrow \quad V_\pi = T(V_\pi)$$

Step 2: We just showed any optimal policy π has a value function that satisfies $V_\pi = T(V_\pi)$, i.e., $V_\pi = V$, the unique fixed point.
Now, consider a policy that is greedy w.r.t. V, i.e., consider $\pi^*(s) = \arg\max_a r(s,a) + - E(V(s') \mid s,a)$

where $E(V(s') \mid s,a)$ is short-hand notation for

$$E(V(s_1) \mid s_0 = s, a_0 = a))$$

Then, $V_{\pi^*} = V$

$\Rightarrow \pi^*$ is optimal

□

Conclusion: $\exists$ a stationary, deterministic π^* which is optimal whose value function satisfies

$$\boxed{V = T(V)}$$

and

$$\boxed{\pi^*(s) \in \arg\max_a r(s,a) + \alpha E(V(s') \mid s,a)}$$

The Q-function:

$$Q(s,a) \triangleq r(s,a) + \alpha E(V(s') \mid s,a)$$

$$\Rightarrow V(s) = \max_a Q(s,a)$$

$$\Rightarrow \boxed{Q(s,a) = r(s,a) + \alpha E\left(\max_{a'} Q(s',a') \,\middle|\, s,a\right)}$$

→ This is another form of the Bellman equation. If we write it as

$$Q = T_Q(Q),$$

one can again show that T_Q is an α-contraction in the $\|\cdot\|_\infty$ norm.

Recall for a fixed memoryless, possibly randomized π

$$V_\pi(s) = r(s,\pi) + \alpha E(V_\pi(s') \mid s)$$

where $E(V_\pi(s') \mid s) = \sum_{s'} V_\pi(s') \sum_a \pi(a \mid s) P(s' \mid s,a)$

Define $Q_\pi(s,a) = r(s,a) + \alpha E(V_\pi(s') \mid s,a)$,

i.e., the reward obtained by taking action a at the current time and the policy π after that.

$$\Rightarrow V_\pi(s) = \sum_a \pi(a \mid s) \left(r(s,a) + \alpha E(V_\pi(s') \mid s,a)\right)$$

$$= \sum_a \pi(a \mid s) Q_\pi(s,a)$$

$$\Rightarrow \boxed{Q_\pi(s,a) = r(s,a) + \alpha E\left(\sum_{a'} \pi(a' \mid s') Q_\pi(s',a') \,\middle|\, s,a\right)}$$

$Q_\pi = T_{Q,\pi}(Q_\pi)$ is also an α-contraction in $\|\cdot\|_\infty$.

By abusing notation, we will use T_π and T for both the V and Q fixed-point equations, but it should be clear from the context.